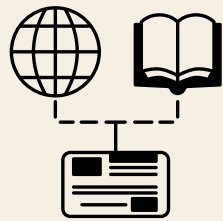


THE PROBLEM



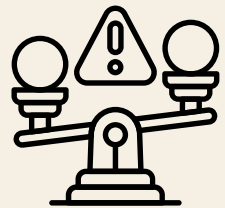
2-4 Weeks

To process ONE loan application



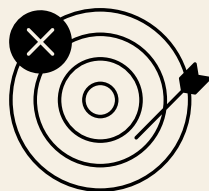
15+ Sources

Must be manually stitched together



High Bias

Subjective, inconsistent analysis



Missed Signals

Early warnings buried in text



Manual Credit Appraisal



Week 1

Collect GST, ITR, bank statements from 5+ portals



Week 2

Read 200-page annual reports, extract financials by hand



Week 3

Google the company, check courts, call references



Week 4

Write the Credit Appraisal Memo in Word. Hope for consistency.

OUR IDEA: CREDEX

End-to-End Credit Underwriting Automation

Automates the full pipeline from document ingestion to credit recommendation.



AI-Generated Credit Appraisal Memo (CAM)

Automatically generates investment-ready credit appraisal reports for committee decisions.



AI Credit Copilot (RAG-Powered)

Conversational assistant enabling analysts to query financial data and risk insights.



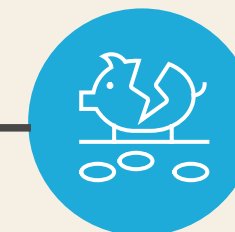
AI Research Agent (Digital Credit Manager)

Automatically gathers news, litigation, ratings, and regulatory intelligence signals.



Portfolio Intelligence Dashboard

Monitors borrowers with real-time risk insights.



Digital Twin Stress Simulator

Simulates financial shocks to predict repayment capacity and default risk.



Explainable Credit Scoring Engine

AI scoring model based on Five Cs of Credit with transparent reasoning.



Cross-Verification & Fraud Detection

AI validates financial data across documents to detect inconsistencies and fraud signals.



Scalable Cloud-Native Architecture

Microservices-based system with parallel processing and horizontal scaling.

SOLUTION WORKFLOW

Credex

WORKFLOW

Dashboard

All entities

1 Entity Onboarding

2 Document Ingestion

3 Data Extraction

4 Analysis & Report

Entity Onboarding

Set up the company profile for credit assessment

1 Company Details

2 Financial Profile

3 Loan Requirements

Company Details

Basic company information

Company Name *

e.g., Tata Steel Limited

CIN (Corporate Identity Number)

e.g., L27100MH1907PLC000260

PAN Number

e.g., AACT2727Q

21-character alphanumeric code

10-character PAN

Industry Sector

Select sector

Date of Incorporation

dd/mm/yyyy

< Back

Next >

Credex

WORKFLOW

Dashboard

All entities

Entity Onboarding

Document Ingestion

Data Extraction

4 Analysis & Report

Document Ingestion

Upload financial documents for Tata Steel Limited. AI will auto-classify each document.

3/4 approved

Proceed to Extraction →

Asset Liability Management

ALM

Approved

Maturity profile, ALM statement

tata_steel_alm_2025.pdf (2 pages) 30% confidence

Shareholding Pattern

SHAREHOLDING

Promoter, FI, DII holdings

Drag & drop PDF or click to browse

Borrowing Profile

BORROWING_PROFILE

Classified

Debt schedule, facility details

tata_steel_borrowing_profile_2025.pdf (2 pages) 99% confidence

Approve

Edit

Annual Report / Financials

ANNUAL_REPORT

Approved

P&L, Balance Sheet, Cash Flow

tata_steel_annual_report_2025.pdf (5 pages) 99% confidence

ACTIVE ENTITY

Tata Steel Limited

WORKFLOW

Dashboard

All entities

Entity Onboarding

Document Ingestion

Data Extraction

4 Analysis & Report

Analysis & Report

Research, cross-verify, score, and generate credit report for Tata Steel Limited

Tata Steel Limited

SCORED

Manufacturing

India

₹ Request: ₹450 Cr (Working Capital)

RISK SCORE

74/100

A

AI RECOMMENDATION

Approve with Monitoring

CIBIL CMR Rank: 3, Willful Defaulter: No, Promoter Pledge: 0.0%

Risk Snapshot

Character

80%

CIBIL CMR Rank: 3, Willful Defaulter: No, Promoter Pledge: 0.0%

Capacity

100%

Interest Coverage: 5.84x, EBITDA Margin: 22.04%, Cash from Ops: ₹39500.0 Cr

Capital

80%

D/E Ratio: 1.12, Net Worth: ₹70800.0 Cr, ROE: 32.63%

Collateral

70%

Total Assets: ₹201000.0 Cr, Current Ratio: 1.15, Contingent Liabilities: ₹4850.0 Cr

Conditions

20%

High-risk findings: 2, Total research findings: 19

Research

Cross-Verify

Credit Score

Stress Test

SWOT

Report

Run Research

News

Tata Steel's Q4 FY24 Performance Impacted by Lower Realizations and ... MEDIUM

Litigation

Arbitration Related to Raw Material Supply Agreements MEDIUM

Credex

WORKFLOW

Dashboard

All entities

Entity Onboarding

Document Ingestion

Data Extraction

4 Analysis & Report

Data Extraction

Extract structured data from uploaded documents

3/5 extracted

Extract All

Proceed to Analysis →

DOCUMENTS

Extracted Data

Schema Config

tata_steel_alm_2025.pdf

ALM

tata_steel_annual_report_2025.pdf

ANNUAL_REPORT

tata_steel_portfolio_performance_2...

PORTFOLIO_PERFORMANCE

tata_steel_borrowing_profile_2025.pdf

BORROWING_PROFILE

tata_steel_shareholding_q4_2025.pdf

SHAREHOLDING

ALM Statement

100% confidence

Edit

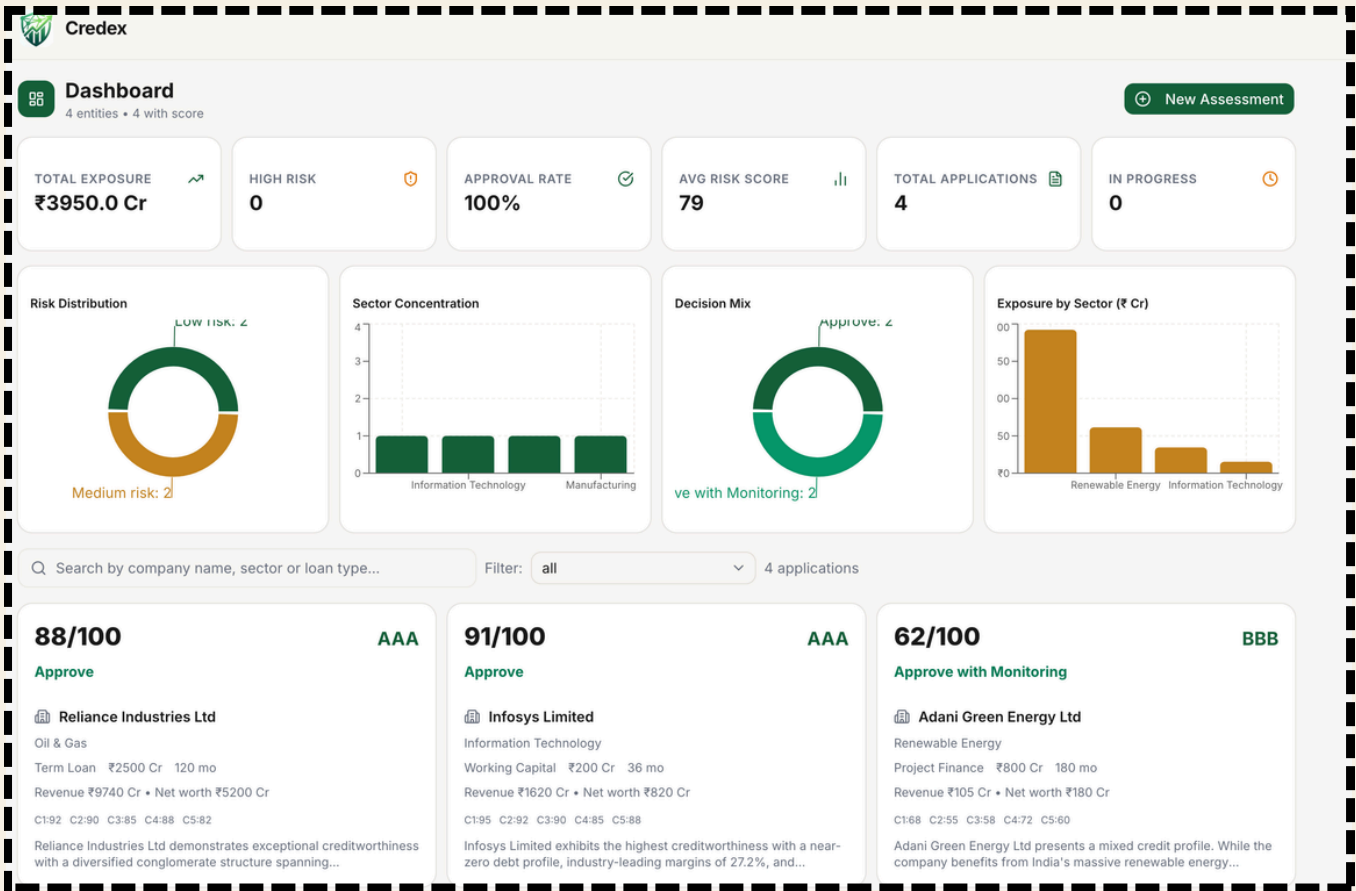
Re-Extract

Reporting Date

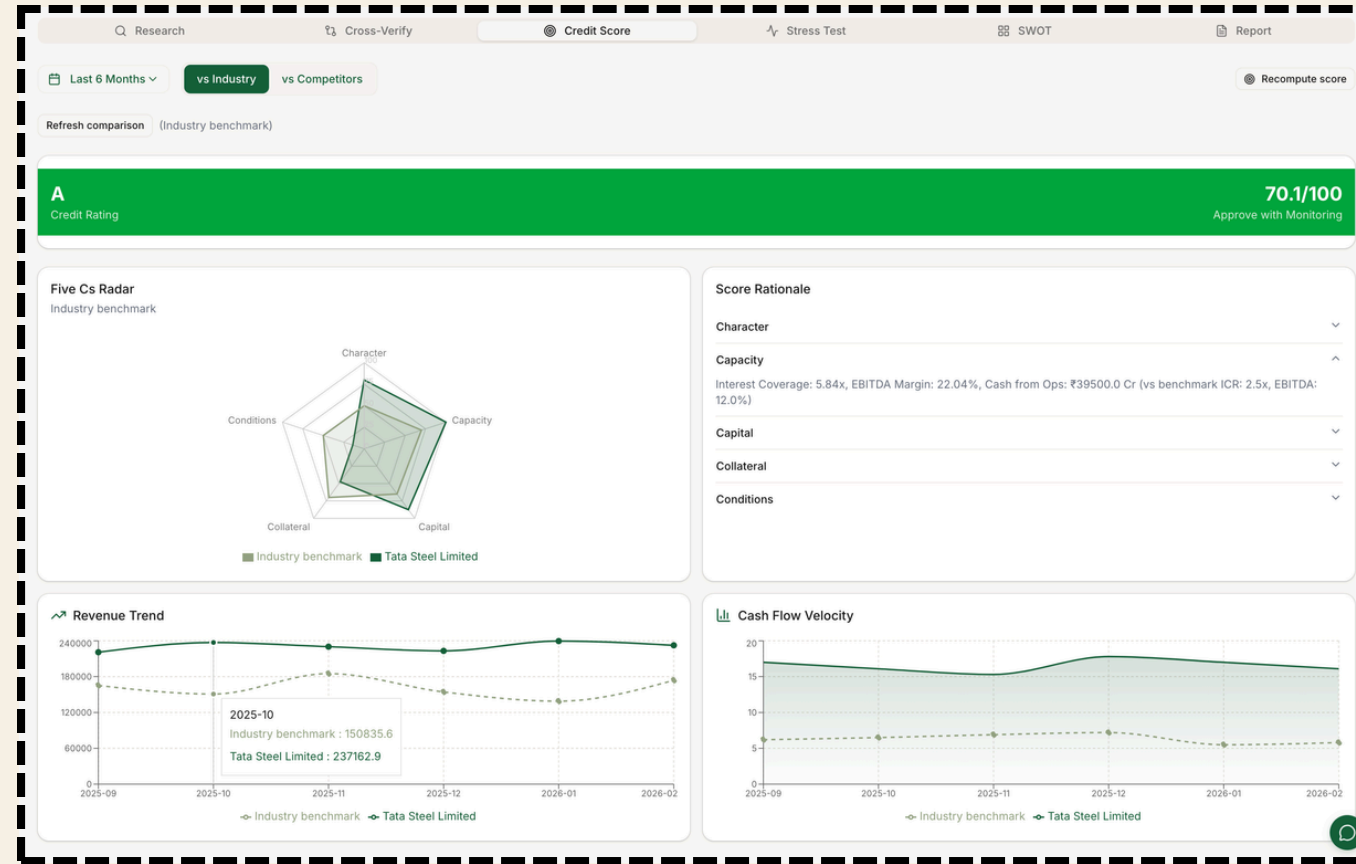
31-Mar-2025

Maturity Buckets

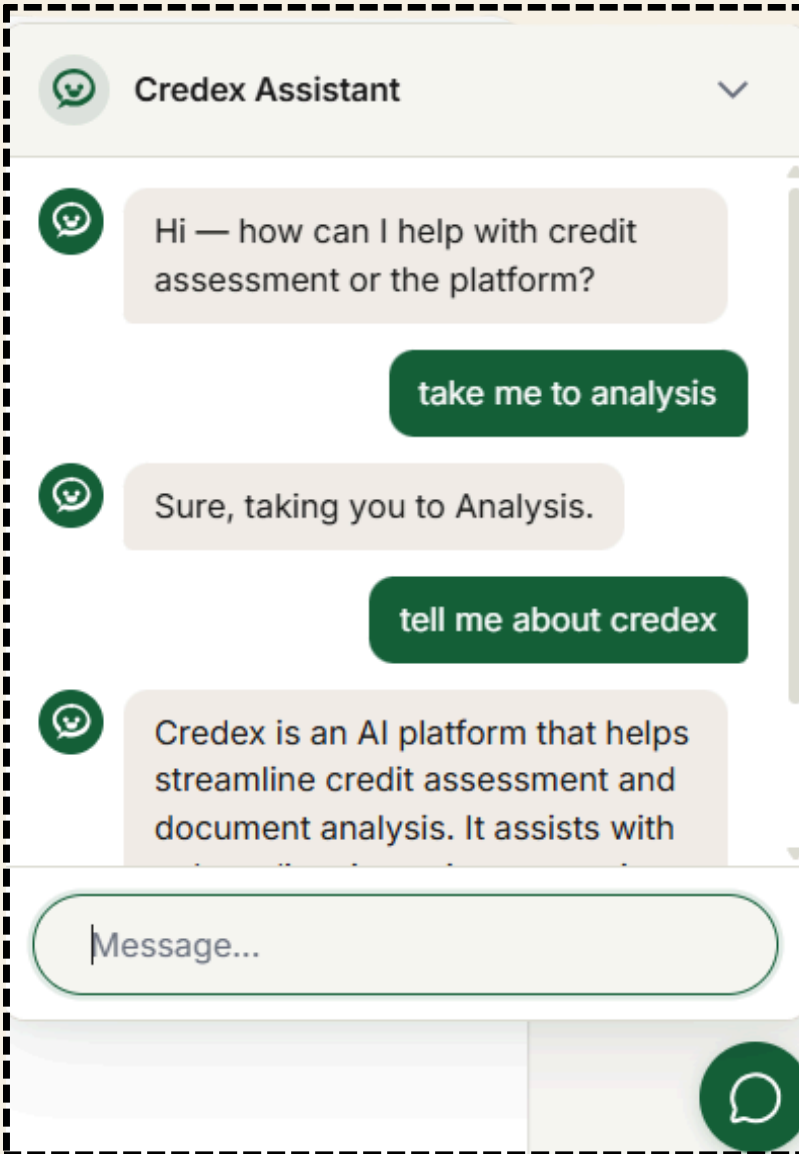
range: 1 day	assets: 2450
liabilities: 3120	gap: -670
range: 2-7 days	assets: 3800
liabilities: 2950	gap: 850
range: 8-14 days	assets: 1920
liabilities: 1450	gap: 470
range: 15-28 days	assets: 4100
liabilities: 3800	gap: 300
range: 29 days - 3 months	assets: 12500
liabilities: 11200	gap: 1300
range: 3-6 months	assets: 8750
liabilities: 7600	gap: 1150



ML-Powered Portfolio Insights



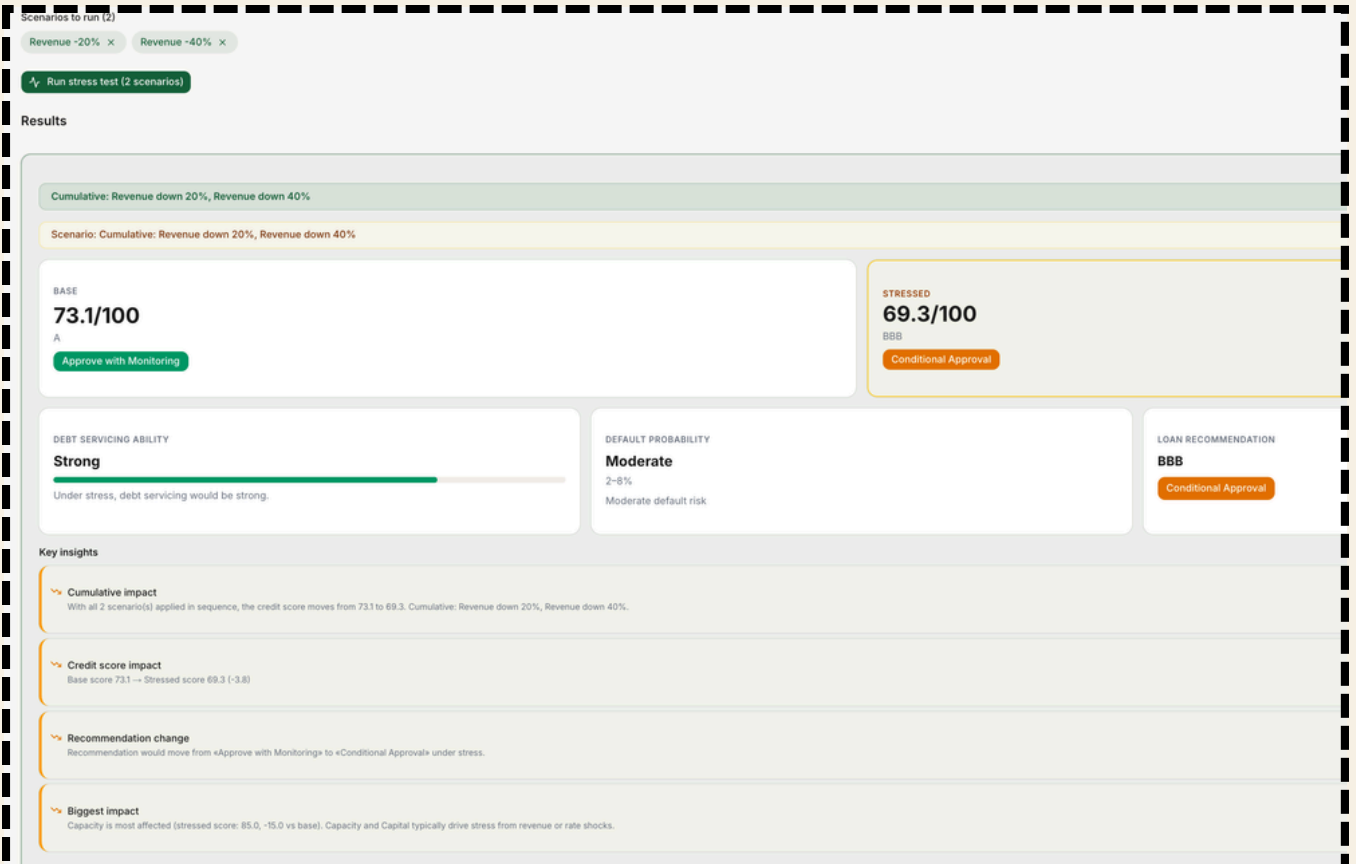
Five-Cs Credit Risk Analysis



AI Copilot



SWOT Analysis



Digital Twin Stress Simulator

System Architecture: Five-Layer Intelligence Pipeline

Linear-yet-iterative pipeline from raw documents to explainable credit decision

L1

UPLOAD

- Parse 4+ document types
- OCR with table detection
- Data pipeline: Bronze → Silver → Gold

TECH

Databricks, Table Transformer
GOT-OCR, LayoutLMv3

L2

RESEARCH

- Research across 8 web domains
- Parallel agent-based execution
- Automated contradiction detection

TECH

LangGraph, Tavily,
SerpAPI
Playwright, FinBERT

L3

ASSESS

- Credit officer review portal
- Site visits and DD notes capture
- Qualitative risk scoring

TECH

React, FastAPI
Gemini NLP scoring

L4

SCORE

- Five Cs credit model (80+ features)
- ML ensemble with SHAP explainability
- LLM-driven qualitative insights

TECH

XGBoost + LightGBM
SHAP explainability

L5

GENERATE

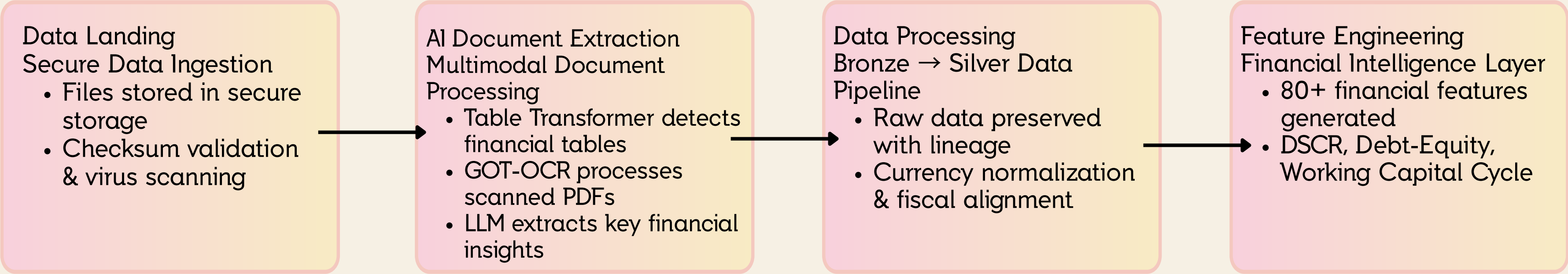
- 12-section CAM report generation
- RBI-mandated report format
- Word and PDF output

TECH

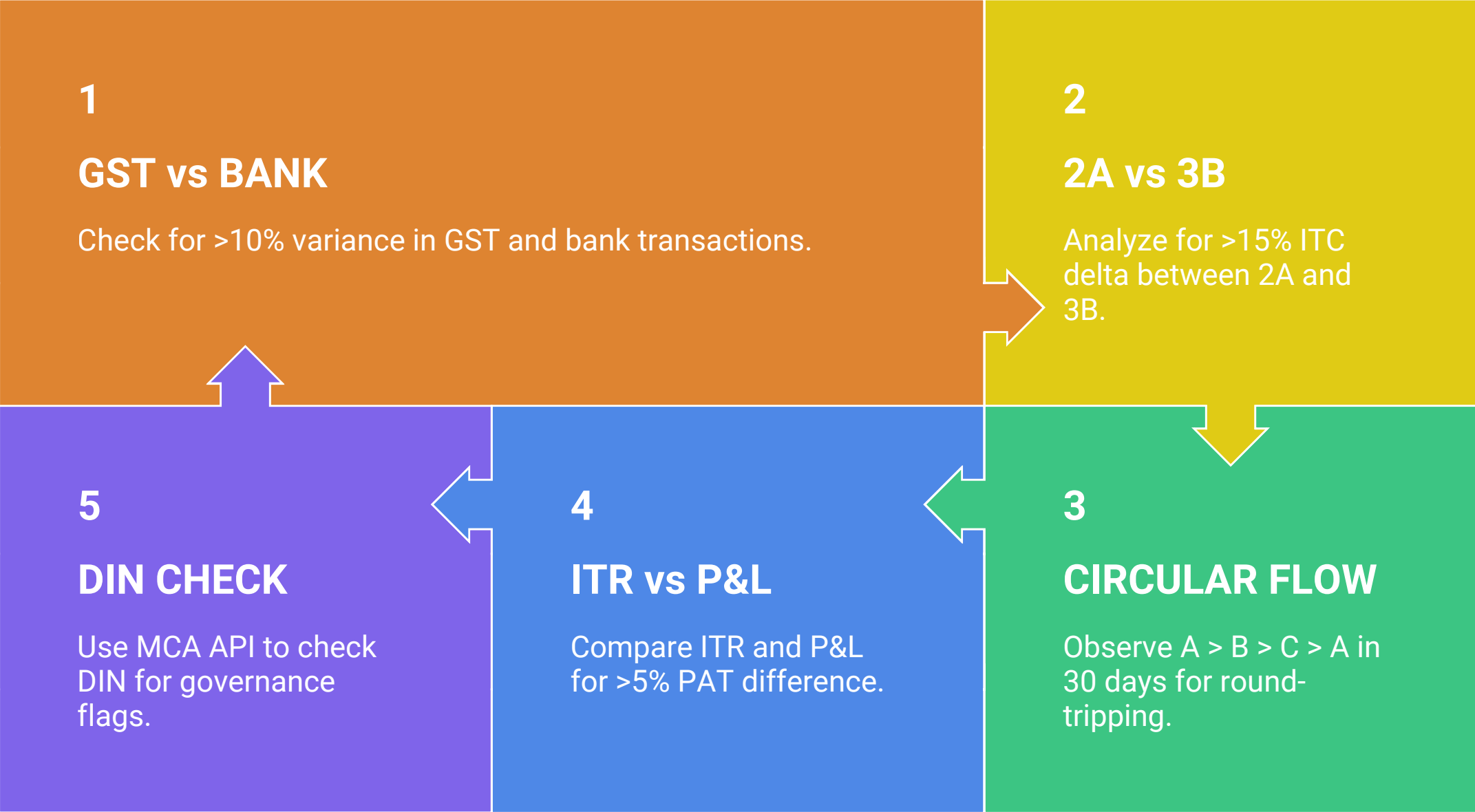
Gemini , python-docx
WeasyPrint, matplotlib



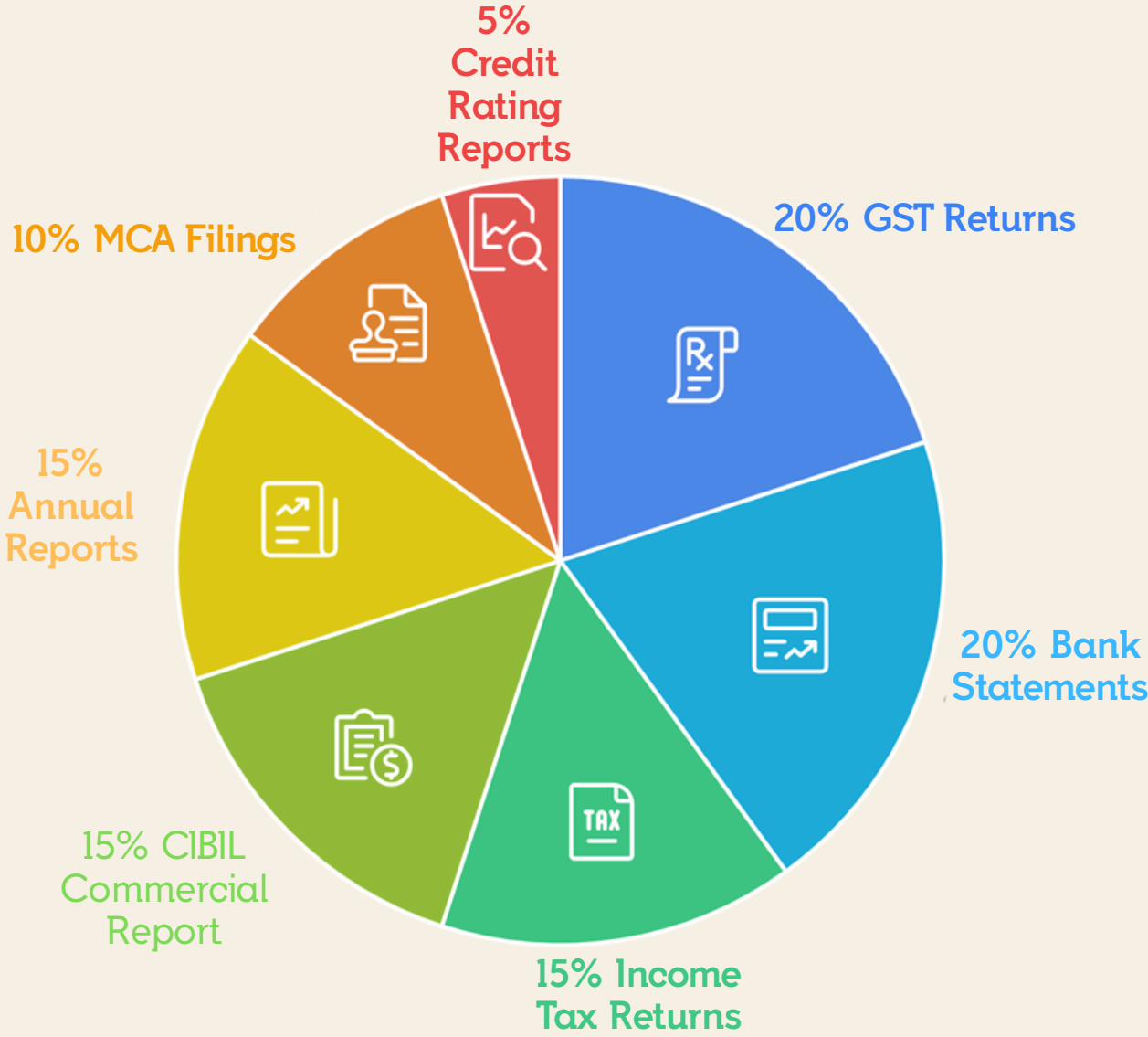
Pillar 1: The Data Ingestor : Reading What Humans Can't — At Scale

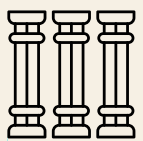


Cycle of Financial Irregularities



DATA SOURCES & SCORE WEIGHTS





Pillar 2: The Research Agent (Digital Credit Manager)

ReAct agent using LangGraph — autonomously plans, executes, and synthesises web-scale research



NEWS

Economic Times, Mint,
Business Standard,
Reuters

Sentiment score
(-1 to +1)

LEGAL

e-Courts, DRT portal,
NCLT/NCLAT orders

Case count, type,
quantum, status

MCA

MCA21 v3 portal:
charges, SH-7, MGT-
7A

Charge details,
pledge status

SECTOR

RBI circulars, DPIIT,
industry body reports

Sector risk rating,
regulatory changes

INSOLVENCY

NCLT IBBI website,
IBC proceedings

IBC petition
filed/admitted?

RATING

CRISIL / ICRA /
CARE
Agency websites

Rating migrations,
outlook changes

Context Extraction

Read Borrower Profile
(CIN, PAN, sector)

Parallel Execution

LangGraph runs 10 concurrent
searches

Contradiction Detection

Compare
web vs provided documents
(e.g., company claims 500 Cr but news
says 200 Cr)

Research Planning

Gemini generates prioritised query
plan

Source Validation

Score credibility

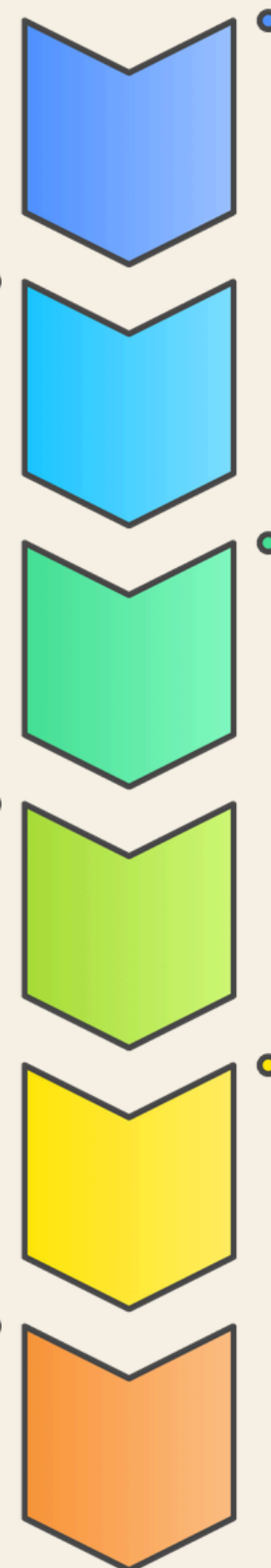
govt > tier-1 media > blog

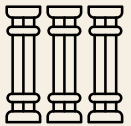
Intelligence Brief

Structured

JSON + human-readable CAM
section

Sentiment: ProsusAI/finbert (6M DL)
| NER: finance-ner (HF)





Pillar 3: The Recommendation Engine

From Data to Decision — Scored, Explained, Documented

FIVE Cs CREDIT SCORECARD

C1	CHARACTER 20% DIN status, litigation, CIBIL, news sentiment, promoter pledge	Who is behind this company?
C2	CAPACITY 25% DSCR, EBITDA margin, interest coverage, cash conversion cycle	Can they repay?
C3	CAPITAL 20% D/E ratio, net worth CAGR, tangible NW, equity infusion	How strong is the balance sheet?
C4	COLLATERAL 20% FOIR, LTV, valuation date, SARFAESI enforceability	What secures the loan?
C5	CONDITIONS 15% Sector risk, regulatory headwinds, economic cycle, geography	What's the market saying?

SCORE -> DECISION MAPPING

<40	40-54	55-69	70-84	85-100
B-	BB	BBB	A	AAA
DECLINE	REFER Committee	CONDITIONAL	APPROVE (Enhanced)	APPROVE

4-LAYER EXPLAINABILITY





PROTOTYPE

<https://credex-bharatveer.vercel.app/>

DEMO VIDEO

https://youtu.be/xlBumSocems?si=dw2OIdhzy_-PLbgp

SAMPLE REPORT

<https://docs.google.com/document/d/1LUCuLmPJJ4G1afgswZD79JayxOYxNqVj/editusp=sharing&oid=106985487891917541120&rtpof=true&sd=true>

THANK
YOU

